QUBO Formulation: Database Transaction Serializability Verification

Database Transaction Serializability Verification

Problem Statement. Given a set of N candidate serial execution orders for a collection of database transactions, each candidate order i is associated with a pre-computed violation cost $c_i \geq 0$ that quantifies the number of observed read/write dependency constraints violated by that order. The task is to select exactly one candidate serial order such that the total violation cost is minimized. A violation count of zero indicates that the execution is fully serializable under the supplied constraints.

Decision Variables. Let N denote the number of candidate serial orders. For each candidate $i \in \{0, 1, \ldots, N-1\}$, introduce a binary decision variable $x_i \in \{0, 1\}$. The variable x_i equals 1 if and only if candidate order i is selected, and 0 otherwise. The selection is enforced by the requirement that exactly one variable takes the value 1.

QUBO Derivation. **Step 1 — Original Objective.** The problem is to minimize the total violation cost over the binary selection variables:

$$\min_{x \in \{0,1\}^N} \sum_{i=0}^{N-1} c_i x_i. \tag{1}$$

Step 2 — Constraint Formulation. The requirement that exactly one candidate order is selected is expressed as:

$$\sum_{i=0}^{N-1} x_i = 1. \tag{2}$$

Step 3 — Penalty Term Expansion. We convert the equality constraint into a quadratic penalty term by adding $\lambda \left(\sum_{i=0}^{N-1} x_i - 1 \right)^2$ to the objective, where $\lambda > 0$ is a penalty weight. Expanding the square yields:

$$\lambda\left(\sum_{i=0}^{N-1} x_i - 1\right)^2 = \lambda\left(\sum_{i=0}^{N-1} x_i^2 + \sum_{i=0}^{N-1}\sum_{\substack{j=0\\ j\neq i}}^{N-1} x_i x_j - 2\sum_{i=0}^{N-1} x_i + 1\right). \tag{3}$$

Using the idempotent property of binary variables, $x_i^2 = x_i$, the expression simplifies to:

$$\lambda\left(\sum_{i=0}^{N-1} x_i + \sum_{i\neq j} x_i x_j - 2\sum_{i=0}^{N-1} x_i + 1\right) = \lambda\left(\sum_{i\neq j} x_i x_j - \sum_{i=0}^{N-1} x_i + 1\right). \tag{4}$$

Recognizing that $\sum_{i\neq j} x_i x_j = 2\sum_{0\leq i<j\leq N-1} x_i x_j$, the penalty contribution to the Hamiltonian becomes:

$$\lambda\left(2\sum_{0\leq i<j\leq N-1} x_i x_j - \sum_{i=0}^{N-1} x_i + 1\right). \tag{5}$$

Step 4 — Combined Hamiltonian. Adding the objective and the expanded penalty term gives the full QUBO Hamiltonian $H(x)$:

$$H(x) = \sum_{i=0}^{N-1} c_i x_i + \lambda\left(2\sum_{0\leq i<j\leq N-1} x_i x_j - \sum_{i=0}^{N-1} x_i + 1\right). \tag{6}$$

Grouping the linear terms results in:

$$H(x) = \sum_{i=0}^{N-1}(c_i - \lambda)x_i + 2\lambda\sum_{0\leq i<j\leq N-1} x_i x_j + \lambda. \tag{7}$$

Step 5 — Q Matrix and Offset. Comparing $H(x)$ with the standard QUBO form $H(x) = \sum_{i\leq j} Q_{i,j} x_i x_j + c$, we read off the general closed-form expressions for the Q matrix entries and the constant offset:

$$Q_{i,i} = c_i - \lambda, \qquad \forall i \in \{0,\ldots,N-1\}, \tag{8}$$

$$Q_{i,j} = \lambda, \qquad \forall 0 \leq i < j \leq N-1, \tag{9}$$

$$c = \lambda. \tag{10}$$

Penalty Weight Justification. Let $C_{\max} = \max_i c_i$ denote the maximum violation cost among all candidates. If the constraint $\sum_{i=0}^{N-1} x_i = 1$ is violated, the squared term $\left(\sum_{i=0}^{N-1} x_i - 1\right)^2$ evaluates to at least 1 for any integer assignment, incurring a penalty of at least λ. The maximum possible reduction in the objective function by selecting a candidate is $C_{\max}$. Therefore, choosing $\lambda > C_{\max}$ ensures that the energy cost of violating the constraint strictly exceeds any potential gain in the objective function, guaranteeing that infeasible configurations are energetically unfavorable.

Correctness Argument. Minimizing $H(x)$ over $\{0,1\}^N$ recovers the optimal serial order because (i) for any feasible assignment where exactly one $x_i = 1$, the penalty term vanishes and $H(x)$ reduces exactly to the original objective $\sum_{i=0}^{N-1} c_i x_i$. (ii) For any infeasible assignment where $\sum_{i=0}^{N-1} x_i \neq 1$, the penalty term adds at least λ to the energy. By enforcing $\lambda > C_{\max}$, the energy of every infeasible configuration is strictly greater than the energy of the best feasible configuration. Consequently, the global minimum of $H(x)$ must lie in the feasible subspace and correspond to the candidate order with the minimum violation cost.

Illustrative Example. Consider an instance with $N = 3$ candidate orders and violation costs $c_0 = 5$, $c_1 = 2$, and $c_2 = 8$. We select $\lambda = 10$, which satisfies $\lambda > \max\{5, 2, 8\}$. The concrete objective is $5x_0 + 2x_1 + 8x_2$. The constraint is $x_0 + x_1 + x_2 = 1$. Substituting into the expanded penalty formula yields:

$$10\left(2x_0x_1 + 2x_0x_2 + 2x_1x_2 - x_0 - x_1 - x_2 + 1\right). \tag{11}$$

Combining with the objective gives the concrete Hamiltonian:

$$H(x) = -5x_0 - 8x_1 - 2x_2 + 20x_0x_1 + 20x_0x_2 + 20x_1x_2 + 10. \tag{12}$$

The corresponding Q matrix entries and offset are:

$$Q_{0,0} = -5, \quad Q_{1,1} = -8, \quad Q_{2,2} = -2, \quad Q_{0,1} = Q_{0,2} = Q_{1,2} = 10, \quad c = 10. \tag{13}$$

Evaluating $H(x)$ over the three feasible binary vectors:

$$x = (1,0,0) \implies H = 5, \quad x = (0,1,0) \implies H = 2, \quad x = (0,0,1) \implies H = 8. \tag{14}$$

The global minimum occurs at $x^* = (0,1,0)$ with energy $H(x^*) = 2$, correctly identifying candidate order 1 as the optimal selection with the minimum violation count.